Fabian **Schuhmann**

Working on problems beyond standard methods

About me

I work on complex systems that do not come with predefined solutions, often under conditions where requirements are incomplete or conflicting. With a background in physics and mathematics, I develop methods and software as problems demand and structure heterogeneous data so that results remain usable beyond a single field. This work takes place in collaborative environments and includes coordinating and supporting contributions from researchers and developers with different areas of expertise.

Specializations

Complex computational systems • Simulation systems • Method and workflow design • High-performance computing

Contact

Fabian Schuhmann
born 07.05.91 in Freising, Germany

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Interests

Video games, Ships, Bouldering, deck-building card games

Languages

German	C2	native
English	C2	● ● ● ●
Danish	A2	● ○ ○ ○

Education & Job experience

2023–today
Post-doc researcher
· Niels Bohr Institute, University of Copenhagen
Designed and maintained computational pipelines for large-scale molecular simulation systems. Developed and integrated custom software components under Linux, released open-source. Provided day-to-day guidance to PhD students and collaborators working on simulation and quantitative analysis tasks.

2020–2023
Doctor rer. nat. (PhD) (Physics)
Summa cum laude · Carl von Ossietzy University Oldenburg
Conducted independent research on complex molecular systems using computational modelling and streamlined analysis methods. Served as the primary developer of a web-based simulation platform, responsible for its ongoing development within the research group.
Title of the thesis: Spotting The Difference - Tailored Tools to Analyze Protein Dynamics

2017–2019
Master of Science. Mathematics
· Ruhr University Bochum
Title of the thesis: Dynamics of billiards with emphasis on the length of the shortest periodic billiard trajectory

2013–2017
Bachelor of Science Mathematics
· Ruhr University Bochum
Title of the thesis (translated): The Brunn-Minkowski theorem and the isoperimetric inequality

2011–2016
Bachelor of Science Management & Economics
· Ruhr University Bochum
Title of the thesis (translated): Global environmental protection as a problem of private provision of a public good

Publications

22 peer-reviewed publications. Selected examples illustrate open-source software development, interdisciplinary data integration, and methodological innovation:

2025 TS2CG as a Membrane Builder, *Journal of Chemical Theory and Computation*
2025 Integrative Molecular Dynamics Simulations Untangle Cross-Linking Data to Unveil Mitochondrial Protein Distribution, *Angewandte Chemie International Edition*
2024 Physics-Based Protein Networks Might Recover Effectful Mutations–a Case Study on Cathepsin G, *Journal of Physical Chemistry B*

IT-knowledge

- Python
- bash
- C++
- Linux
- HPC
- Windows
- Containers
- Git

Competitive Grants & Awards

2025 Grant from the Dr. phil. Ragna Rask-Nielsens Grundforskningsfond
2024 ESMTB Travel Award for the ECMTB 2024
2023 ESMTB Travel Award & SIAM Travel Award for the SIAM CSE 2023
2023 SIAM Hackathon Finalist
2022 Grant from the German Academic Exchange Service (DAAD)